Grey Box: Replicating Agents' Hidden States from Their Transcripts, to Predict and Prevent the Violations of Closed Agents

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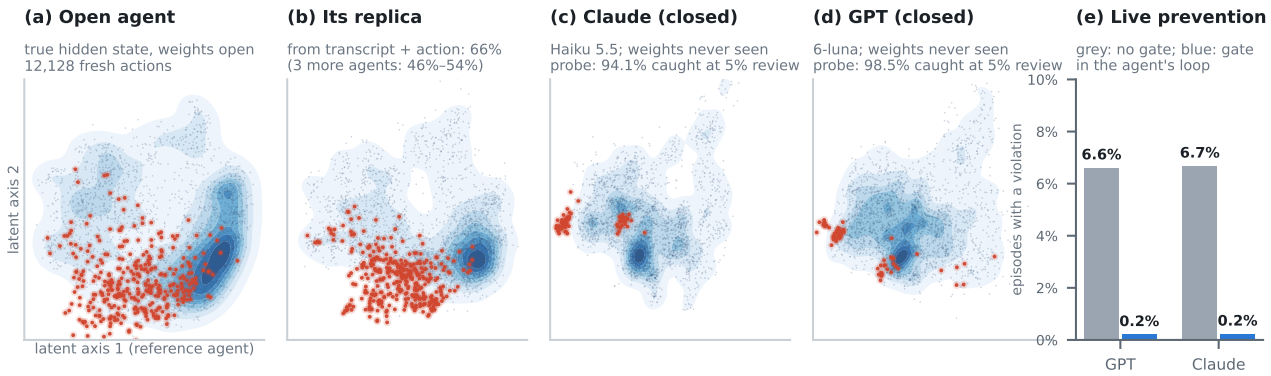


Figure 1. **Seeing inside closed agents.** All panels share the two main axes of one reference open agent's hidden state (Gemma-4, at the moment before it acts); blue: density of safe actions, red: policy violations. (a) The open agent's true hidden state on 12,128 fresh actions. (b) Its grey-box replica, computed from the transcript and the proposed action only (Section 4.2). (c, d) Actions of two closed agents, mapped through the same reader. Their own coordinates are not identifiable (Proposition 1); we never saw their weights and fit nothing on them. A probe on the reader's state catches 94% and 98% of their violations when 5% of actions go to review. (e) In the agents' tool loops, blocking the flagged actions cuts the episodes with an executed violation from 6.6–6.7% to 0.2% (Section 4.5).

Abstract

Hidden-state probes can stop an agent's unsafe actions, but closed agents do not share their weights, and hidden coordinates cannot be recovered from outputs (a gauge symmetry). We show that an open model that reads the same transcript can instead *replicate* the state: across four open agent architectures it reconstructs 46–66% of the variance of the main hidden-state components, more than text features do, and white-box probes score at least as well on the replica as on the true state, with the lower convex losses that a Rao–Blackwell argument predicts. Because the transcript and the proposed action are Bayes-sufficient for the policy labels they decide, a probe on the reader, trained only on an open agent, catches 94% and 98% of two closed agents' violations at a 5% label-free review budget on the benchmark it was trained on, 89% at the same budget when the agent is told it is monitored, and cuts episodes with an executed violation from about 7% to 0.2% in the agents' live tool loops. All confirmatory tests were preregistered and hash-locked, and we release an SDK, a server and a browser tool to train and audit such monitors.

1. Introduction

Interpretability reads what a model knows and what it will do from its hidden state (Burns et al., 2023; Marks & Tegmark, 2023; Zou et al., 2023; Arditi et al., 2024; Goldowsky-Dill et al., 2025). For agents that act through tools, a probe on the hidden state forecasts an unauthorised action before it runs, and a gate can stop it (Fayyazi et al., 2026). These tools need the weights. The most capable agents are closed: their providers return sampled text, at most some log probabilities, and a summary of the reasoning.

The grey box. We ask what an outside observer can recover of a closed agent's hidden state, and whether it is enough to predict and prevent the agent's violations. Two facts answer it (Section 2). First, hidden coordinates are not identifiable from outputs: any reparametrisation of the hidden space that the next layers undo leaves every output unchanged, a gauge symmetry (Proposition 1). Second, the hidden state is a

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deterministic function of the tokens that the agent read. An open model that reads the same tokens can therefore hold a *replica* of it: the conditional expectation of the agent's state given the reader's state (Proposition 4). For a policy check of a proposed action, nothing is lost by reading the transcript instead of the weights (Corollary 1).

Contributions.

1. *Theory* (Section 2): gauge non-identifiability; two gauge-invariant objects, the behavioural state (read in Fisher–Rao coordinates by a battery designed from black-box answers) and the token-determined state; the replica as a conditional mean that Rao–Blackwellises white-box probes; Bayes sufficiency of the proposed action, with a ceiling on any monitor's robustness to rewording; and label-free decision rules with exact coverage and bounds under shift and attack bursts (Algorithms 1 and 2).
2. *Replication* (Section 4.2): one reader replicates the decision state of four agent architectures on unseen scenarios, and white-box probes score at least as well on the replica as on the true state, in all four.
3. *Prediction and prevention on closed agents* (Sections 4.4 and 4.5): a probe trained only on an open agent catches 94% and 98% of two closed agents' violations and blocks them live; it keeps catching them when the agent knows it is monitored, and a cut set again on the new traffic restores the 5% review budget live (Section 4.7); labels of a new policy collected on open agents transfer to a closed agent (Section 4.6).
4. *A released tool* (Section 5): an SDK, a server and a browser trainer, with Python and browser implementations that give near-identical models.

Every confirmatory test was preregistered: its document and analysis code were hash-locked (SHA-256) before the data existed, and for the later tests the lock was also published first. All tests, including those that failed or could not be done, and the limits of each claim, are in Sections A and H.

2. Method

2.1. Setting and threat model

An agent is an autoregressive policy π_θ over tokens. At step t it has read and written the tokens Z_t. A deployment sees V_t, and H_t is the rest, so $Z_t = (V_t, H_t)$; with reasoning off, as in our closed-agent tests, $H_t = \varnothing$. Z_t^- are the tokens before the action, and V_t is a function of Z_t^-. The agent samples an action $A_t \sim \pi_\theta(\cdot \mid Z_t^-)$ with noise ξ_t that is independent of everything else. Its state at the decision token is $\phi_t = \Phi_\theta(Z_t) \in \mathbb{R}^d$, a deterministic function of the tokens. W are facts of the world that the agent has not observed, and $Y \in \{0, 1\}$ marks a policy violation. A *monitor* maps what it sees to a score s_t and escalates the action when $s_t \geq c$; its *budget* β is the share of actions that it escalates.

Threat model. The operator has V_t and the proposed action A_t; it never has θ, ϕ_t or H_t. The monitor must decide before A_t runs. Labels exist for open agents, not for the monitored agent: its traffic is unlabelled, and the scale of a score does not transfer between agents. The agent may know that it is monitored (Section 4.7).

2.2. What cannot be read: a gauge symmetry

Proposition 1 (gauge)**.** *Take a pre-norm residual network with RMS normalisation, with the normalisation gains folded into the adjacent weights. For every orthogonal G on the residual stream, the network whose embedding, write matrices and write biases are multiplied by G on the left, and whose read matrices and unembedding are multiplied by $G^\top$ on the right, has hidden states $G\phi$ and the same output distribution on every input. At a layer that is read only through linear maps, the same holds for every invertible G, with G^{-1} in place of $G^\top$.*

Thus no estimator that uses only outputs can recover hidden coordinates beyond this group, and a probe direction w on one parametrisation is $G^{-\top}w$ on the other. Post-block normalisation restricts G to the maps that commute with its gains, and LayerNorm to the maps that fix the all-ones vector (Section B). As in a gauge theory, only invariant quantities are observable. Two invariants carry the safety-relevant state: the law of the agent's future outputs (its *behavioural state*) and the token history Z_t itself, of which ϕ_t is a function. We build a grey box from each.

2.3. Reading the behavioural state

The *interrogation state* of a context x for a battery of yes/no questions $Q = (q_1, \ldots, q_m)$ is $\Psi_Q(x) = \big(P_\pi(\textsc{yes} \mid x \oplus q_j)\big)_j$. Each question is asked in a side branch, so the agent's conversation never sees it. Ψ_Q is a vector of probabilities of fixed events, so it is gauge-invariant and comparable across models.

Proposition 2 (Fisher–Rao coordinates)**.** *For one Bernoulli coordinate the map $\vartheta = 2\arcsin\sqrt{p}$ is an isometry from the Fisher–Rao metric $\mathrm{d}p^2/\big(p(1-p)\big)$ to the Euclidean line. With n sampled answers and $np(1-p) \gg 1$, $\mathrm{Var}(\hat{\vartheta}) \approx 1/n$ for every such p, so $\vartheta(\hat{\Psi}) \in [0, \pi]^m$ has isotropic noise of known size.*

We design the battery from black-box answers alone (Algorithm 2): from a pool of candidate questions, add the one that splits the most classes of contexts with equal answers, and stop when no candidate splits a class.

Proposition 3 (battery design)**.** *Algorithm 2 stops with the partition of the contexts that the whole pool induces and asks at most (number of its classes $-\,1$) questions. If the*

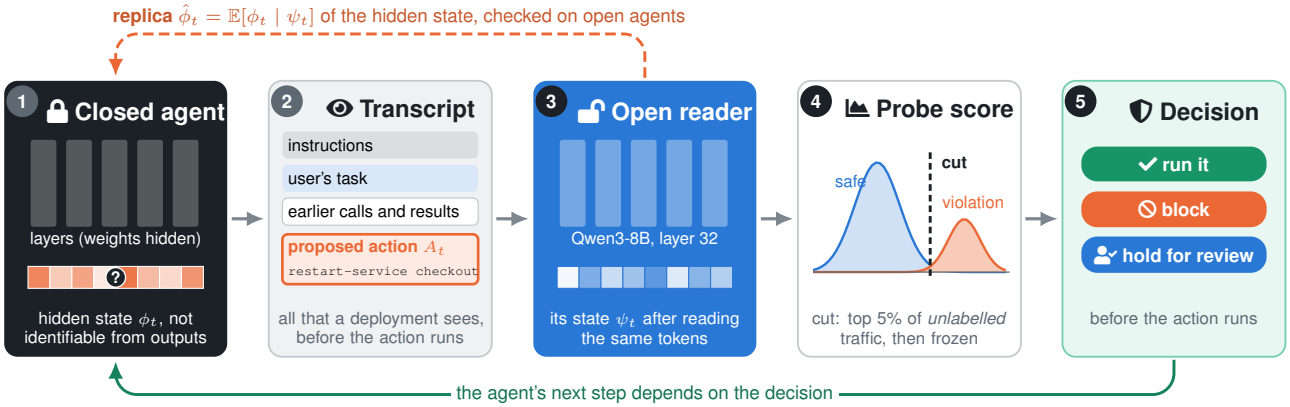


Figure 2. **The grey box, as one causal chain.** (1) A closed agent's weights and hidden state ϕ_t are out of reach, and its hidden coordinates cannot be identified from its outputs (Proposition 1). (2) The operator sees the transcript and the proposed action, before the action runs; when these decide the label, they are sufficient (Corollary 1). (3) An open reader reads the same tokens; its state ψ_t holds a replica $\mathbb{E}[\phi_t \mid \psi_t]$ of the agent's state (orange arrow), checked against the true state of four open agents (Propositions 4 and 5, Section 4.2). (4) A probe on ψ_t, trained only on an open agent, scores the action; the cut is the top 5% of the agent's own unlabelled traffic, then frozen (Propositions 7 to 9). (5) The action runs, is blocked, or is held for a reviewer, and the agent continues from that decision (Algorithm 1).

model's answers come from a finite automaton and the pool holds a distinguishing suffix for every pair of its states, this partition is the minimal (Myhill–Nerode) state, and no black-box method can identify more.

2.4. Replicating the token-determined state

A *reader* is an open model R that reads the same visible tokens and the proposed action; $\psi_t = R(V_t, A_t)$ is its hidden state at a fixed layer.

Proposition 4 (access and replica)**.** *(a)* $\sigma(Z_t, \phi_t) = \sigma(Z_t)$. *For every label* Y*, the best monitor that reads* (Z_t, ϕ_t) *is no better than the best that reads* Z_t*, under every strictly proper score and at every point of the ROC curve. Under log loss, the best monitor that reads* Z_t *beats the best that reads* V_t *by exactly* $I(Y; H_t \mid V_t)$*. (b) The* L^2*-optimal replica* $\hat{\phi} = \mathbb{E}[\phi \mid \psi]$ *satisfies* $\mathrm{Cov}(\hat{\phi}) \preceq \mathrm{Cov}(\phi)$ *and, for every component* k*,* $\mathrm{Var}(\hat{\phi}_k) = R^2_k \mathrm{Var}(\phi_k)$*, with* $R^2_k = 1 - \mathbb{E}(\phi_k - \hat{\phi}_k)^2 / \mathrm{Var}(\phi_k)$*. The same identity holds for the best linear replica, in sample. (c) For every linear readout* w*,* $\mathbb{E}[w^\top \phi \mid \psi] = w^\top \hat{\phi}$*: a probe applied to the replica is the conditional expectation of the same probe on the true state.*

Part (a) says why a replica is possible: with reasoning off, ϕ_t is a function of tokens that an outside model can also read; with hidden reasoning, the replica is the conditional mean given the visible tokens only. Part (b) predicts the shape of the replica, and part (c) says what a white-box readout computes on it. The next result says when that readout is *better* than the original.

Proposition 5 (the replica Rao–Blackwellises a probe)**.** *Let* $s = w^\top \phi$ *be a probe on the agent's state and* $\hat{s} = \mathbb{E}[s \mid \psi]$ *its readout on the* L^2*-optimal replica. If* $Y \perp\!\!\!\perp \phi \mid \psi$*, then* $\mathbb{E}\,\ell(\hat{s}, Y) \le \mathbb{E}\,\ell(s, Y)$ *for every loss* $\ell(s, y)$ *that is convex in* s*. The same holds for every convex pairwise loss* $\mathbb{E}\,\ell(s_i - s_j)$ *of independent violating and safe actions* i, j*, the convex surrogates of the AUROC.*

The premise says that the agent's state holds no label information beyond what the reader holds; Corollary 1 below makes it plausible for policy labels. Under it, the replica removes only variance that is noise for the label. We estimate $\hat{\phi} = M\psi + b$ by ridge regression on paired steps of an open agent, on the top principal components of its state; the proposition applies to this estimate only as far as the linear map approximates the conditional mean.

2.5. The proposed action is sufficient

Corollary 1 (sufficiency of the proposed action)**.** *Let* $Y = g(A_t, V_t, W)$ *with* $W \perp\!\!\!\perp Z_t^- \mid V_t$*. Then* $P(Y \mid Z_t^-, A_t) = P(Y \mid V_t, A_t)$*, and every monitor that reads the agent's internals at or before the action has a Bayes risk at least that of the best monitor that reads* (V_t, A_t)*.*

A probe read before the action must also predict which action the agent will sample; a monitor that reads the action does not. Training a monitor is therefore a supervised problem on (V_t, A_t), and the reader matters through how well a linear probe on its state learns $P(Y \mid V_t, A_t)$ under distribution shift.

An agent or attacker can also reword what an action says without changing what it does. Let $\kappa(A)$ be the decisive part of an action (its tool and targets) and let the free text be the rest.

Proposition 6 (a ceiling under rewording)**.** *Suppose that for every violating action an attacker can reach a still-violating action with the same $\kappa(A)$ and any free text that safe actions with the same $(V, \kappa(A))$ use, and that the attacker picks, among these, the text with the lowest score of the monitor. Then for every monitor s and cut c there is a monitor m of $(V, \kappa(A))$ alone that escalates no more safe actions than s and, on clean traffic, catches at least the share of violations that s catches under the attack.*

Under rewording, no monitor's operating point beats the clean ROC curve of the best monitor of the decisive fields. Free text helps a monitor only on traffic that nobody rewords; a canonical monitor's clean accuracy is achievable under rewording and is a lower bound on the ceiling.

2.6. Label-free decisions at a fixed budget

A score's scale does not transfer between agents (Section 4.4), so the cut must come from the monitored agent's own traffic, without labels (Algorithm 1).

Proposition 7 (a frozen cut)**.** *Let the scores of n unlabelled warm-up actions and of a new action be exchangeable with no ties, and let $\hat{c}$ be the k-th smallest warm-up score. Then $P(S_{n+1} > \hat{c}) = (n+1-k)/(n+1)$, which for $k = \lceil (1-\beta)(n+1) \rceil$ lies in $\left(\beta - \frac{1}{n+1}, \beta\right]$. For i.i.d. scores with continuous distribution function F, the long-run escalation rate $1 - F(\hat{c})$, over warm-up draws, follows* $\mathrm{Beta}(n+1-k, k)$*. The interpolated empirical $(1-\beta)$-quantile lies between two adjacent order statistics, so its escalation probability is within $\frac{2}{n+1}$ of β.*

With $n = 500$ and $\beta = 0.05$ the escalation rate of the order-statistic cut has mean 0.050 and standard deviation 0.0097 (the interpolated cut's mean is between 0.050 and 0.052).

Proposition 8 (shift)**.** *If deployment scores follow Q and the warm-up scores follow P, then for every cut c, $|Q(S > c) - P(S > c)| \le K(P, Q) = \sup_c |F_P(c) - F_Q(c)|$. A frozen cut therefore escalates within $K(P, Q) + \frac{2}{n+1}$ of β, and its share of violations caught moves by at most the Kolmogorov distance between the laws of the violations' scores.*

Proposition 9 (bursts)**.** *Let the score laws of violations and of safe actions not depend on ρ. A cut set at the top β of a window in which a share $\rho > \beta$ of the actions are violations catches at most β/ρ of them. A frozen cut catches each violation with probability $P(S \ge \hat{c} \mid Y = 1)$, whatever ρ is.*

A cut that keeps moving with recent traffic thus fails exactly when an attack arrives, while a frozen cut keeps its rate; under a shift of the safe traffic, the frozen cut must be set again (Proposition 8).

Metrics. We report the share of violations caught when *exactly* β of the actions, the highest-scored, are escalated (actions tied at the cut count in proportion), and the AUROC within strata (template, tool or suite), $\sum_k n_k^+ n_k^- \, \mathrm{AUC}_k / \sum_k n_k^+ n_k^-$, which removes what the stratum alone explains. Intervals are 95% percentile intervals from a bootstrap over scenarios or episodes. Proofs of all statements are in Section B.

Algorithm 1 The grey-box monitor

Require: open-agent steps $\{(V_i, A_i, Y_i)\}$; reader R, layer ℓ; budget β; warm-up length n

$x_i \leftarrow$ render(V_i, A_i): instructions, task, earlier calls and results, proposed action
$\psi_i \leftarrow R_\ell(x_i)$ at the last token {the reader's state}
fit f: standardise $\to$ PCA(256) $\to$ logistic (C=0.05) on (ψ_i, Y_i) {never on the test agent}
for the first n actions A_t of the monitored agent **do**
  record $s_t \leftarrow f(R_\ell(\text{render}(V_t, A_t)))$; run A_t {no labels}
end for
$\hat{c} \leftarrow$ the $(1-\beta)$-quantile of $s_{1..n}$; freeze it {Propositions 7 and 9}
for each later proposed action A_t **do**
  if $f(R_\ell(\text{render}(V_t, A_t))) \ge \hat{c}$ **then** block A_t, or hold it for review **else** run A_t
end for

3. Experimental setup

Agents and data. CAPE-Bench (Fayyazi et al., 2026) gives matched tool-use tasks in which one fact decides whether an action is authorised; an environment checker labels each action. Open agents: Gemma-4 (26B), gpt-oss-20b, Qwen3.8-27B and Muse-Glimmer-30B. Closed agents, through their public tool-use APIs with reasoning off: Claude Haiku 5.5 and GPT-6-luna. External benchmark: AgentDojo prompt injections (Debenedetti et al., 2024). **Reader and monitor.** The reader is Qwen3-8B; it reads the agent's instructions, the user's task, the earlier tool calls with their results and the proposed action, and ψ_t is its residual stream at layer 32 at the last token. The monitor is a logistic probe on ψ_t (standardisation, 256 principal components, $C = 0.05$), trained only on 17,623 Gemma-4 CAPE-Bench steps and never fit on a test agent unless stated. **Baselines.** The agent's own internals (where open), TF-IDF with the reader's input and labels, a zero-shot 70B judge (Llama-3.3-70B, the same input) and tool identity (the action's tool alone).

4. Results

4.1. Reading the behavioural state

On models with known internals, the interrogation state recovers the state. A battery of 21 questions, chosen from black-box answers only, separates all 21 reachable states (and a dead state) of a program compiled into a transformer; after a random rotation of the residual stream, a readout

Table 1. **Detecting policy violations before the action runs**, as the distribution moves away from training. AUROC within template (within suite on AgentDojo), so that what the tool or task alone predicts does not count. The judge and TF-IDF see the same transcript and action as the reader; the agent's internals need its weights. Nothing is fit on a test agent. Best in each row in bold, second underlined. Last column: the preregistered contrast of the row (reader − internals, I; non-inferiority to internals, NI; reader − judge, J; the same within tool, J_{tool}), with its 95% interval from a bootstrap over scenarios or episodes; each met its criterion.

Setting	Agent	Actions	Monitor				Preregistered contrast
			Agent's internals	70B judge (zero-shot)	TF-IDF (same labels)	**Grey-box reader**	
Open agent, the training distribution							
Same scenarios	Gemma-4	12,128	0.979	0.910	<u>0.992</u>	**0.994**	I $+0.015$ [0.007, 0.026]
Unseen task families	Gemma-4	12,128	**0.976**	0.910	0.950	<u>0.969</u>	NI -0.007 [-0.014, 0.000]
Another architecture	gpt-oss-20b	5,229	–	0.833	**0.943**	<u>0.915</u>	J $+0.082$ [0.043, 0.129]
Closed agents, through their APIs (reasoning off)							
CAPE-Bench	Claude Haiku 5.5	3,128	–	0.633	**0.997**	<u>0.991</u>	J $+0.360$ [0.302, 0.413]
CAPE-Bench	GPT-6-luna	3,627	–	0.942	<u>0.968</u>	**0.995**	J_{tool} $+0.039$ [0.015, 0.066]
A new benchmark: AgentDojo prompt injection, a policy absent from training							
No AgentDojo labels	two open	3,455	–	<u>0.537</u>	0.503	**0.781**	J $+0.249$ [0.136, 0.369]
Labels of open agents only	GPT-6-luna	3,925	–	<u>0.787</u>	–	**0.922**	J $+0.136$ [0.078, 0.191]

that uses the known layout falls from $r = 0.9995$ to -0.46 while every answer is unchanged, as Proposition 1 requires. On Othello-GPT, 61 questions recover the board-carrying hidden state at R^2 0.967 on 1,000 fresh games (input-only 0.873). On eight open chat models the answers recover the hidden state better than a text embedding ($+0.120$ R^2, 8 of 8 models), and readouts trained only on open models transfer to Claude (harmful request 0.993, refusal 0.997 AUROC).

4.2. Replicating the hidden state

On Gemma-4, where the true state is known, we fit the map from the reader's state to the agent's decision-point state (five layers, top 128 principal components, 80.5% of the variance) on development steps. On 12,128 fresh steps the replica reconstructs 66.4% of the variance from the transcript alone, against 58.6% for TF-IDF features ($+0.079$ [0.075, 0.083]), and it is better on every one of the 128 components (Figure 3a). A probe trained on the agent's *true* states scores 0.984 on true fresh states and 0.987 on the replica ($+0.003$ [-0.002, $+0.007$]). The same reader replicates three more architectures, a mixture of experts (gpt-oss-20b), hybrid linear attention (Qwen3.8-27B) and a dense model (Muse-Glimmer-30B), on scenarios that the map never saw: R^2 0.494, 0.542 and 0.462, each above text features ($+0.102$, $+0.101$, $+0.061$, all intervals above 0), and the true-state probe works on every replica (Figure 3b,c). Before reading an agent's state we checked that we reproduce its context: teacher-forced, its own model predicts 91–100% of the next 16 tokens of its actual tool call. The theory predicts the shape of the replica. Across the 128 components, the ratio of the replica's variance to the state's variance closely tracks the component's R^2 (Figure 3d; correlation 0.99, slope 0.89 on fresh steps), the identity of Proposition 4(b).

The replica improves the white-box probe. In all four agents the probe trained on true states scores *higher* on the replica than on the true states, by pairwise and log losses and by AUROC (Table 5 in Section F; exploratory analysis of the preregistered data). Proposition 5 predicts the lower convex losses when $Y \perp\!\!\!\perp \phi \mid \psi$, and the data are consistent with that premise: the part of the probe score that the replica misses (23–33% of its variance) adds at most 0.0006 AUROC to the replica score, and the pairwise logistic risk falls by 19–61%. A TF-IDF replica gains as much on the three new agents (Section A): the gain comes from shrinkage toward a conditional mean, not from the reader alone.

4.3. The transcript is enough

Corollary 1 says that the agent's internals cannot beat a monitor that reads the transcript and the action. A probe on the agent's own internals scores 0.841 at the start of the turn and 0.979 after its reasoning: the white-box advantage over turn-start monitors is *when* it reads. A reader probe trained on the same labels scores 0.994, above the agent's internals ($+0.015$ [0.007, 0.026]), and it is non-inferior on unseen task families (0.969 vs 0.976). In an exploratory check, adding the agent's hidden reasoning to the reader's input, given the action, changes the AUROC by -0.0014 [-0.0046, $+0.0014$]: given the action, the hidden reasoning adds no measurable information, as Proposition 4(a) allows. What the reader adds over word counts is robustness to shift (Table 1): TF-IDF with the same labels equals the reader on the training benchmark, but it is lower on unseen task families (0.950 vs 0.969) and at chance on AgentDojo (0.503), where the reader keeps 0.781 against a zero-shot judge's 0.537.

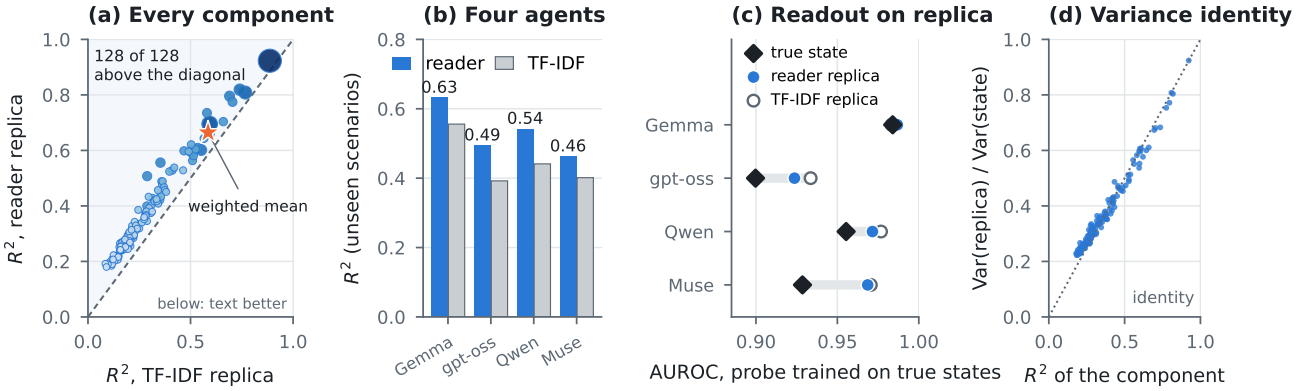


Figure 3. Replicating an agent's hidden state from its transcript. (a) Each of the 128 principal components of Gemma-4's decision state (marker area: its share of the state's variance): the share of its variance that the reader replicates on 12,128 fresh steps, against TF-IDF features on the same steps. The reader is better on all 128; star, the variance-weighted mean, 0.664 against 0.586 (+0.079 [0.075, 0.083], preregistered). (b) predicts for a conditional mean. (b) Variance-weighted R^2 on unseen scenarios for four architectures. (c) A probe trained on the agent's *true* states, applied to the true states (diamond), the reader replica (blue) and the TF-IDF replica (ring); within-template AUROC. (d) Per component, the replica's variance relative to the state's closely tracks the replicated share, as Proposition 4(b) predicts for a conditional mean.

4.4. Two closed agents

Nothing was fit on the closed agents (Table 2). On Claude Haiku 5.5 the monitor catches 94.1% of 68 violations when 5% of the actions go to review, with a cut set without labels; the zero-shot judge catches none. One family that is absent from training makes up 62% and 59% of the two agents' violations; without it, the monitor catches 92.3% and 96.3% (exploratory). On GPT-6-luna all three preregistered endpoints pass: pooled AUROC 0.996 vs the judge's 0.778 (+0.217 [0.174, 0.265]), beyond tool identity (+0.141 [0.111, 0.171]), and within tool (+0.039 [0.015, 0.066]). The scale of the scores does not transfer between agents (on Claude a cut at probability 0.5 catches 3%), but the ranking does. In an exploratory replay, a cut at the top 5% of the agent's own unlabelled traffic, frozen after 500 actions, catches 95.4% online on Claude and 98.4% on GPT. Under bursts of attacks in which half the actions are violations it stays at 94%, while a cut that keeps moving over a window of 5,000 actions falls to 36% (over 500: 20%) (Figure 4a, Figure 5c): the frozen cut is unaffected, as Proposition 9 predicts, and the moving cut loses most of the burst.

4.5. Live prevention

We ran the monitor as a server in the tool loop of each closed agent, on the same scenarios with and without the gate; the cut came from the agent's earlier unlabelled traffic, and a blocked action does not run. In all four preregistered live tests the gate removed nearly all executed violations (Table 3), and neither agent executed a violation after a block (Figure 5a). Holding an escalated action for a reviewer kept the safety gain; in an exploratory comparison of runs that did not take place at the same time it reached more goals than blocking (Figure 5b), with the environment's oracle as an upper bound on a human reviewer, and the preregistered bound on its goal cost was not shown (Section A). The round trip was 89 ms on GPT-6-luna and 80 ms on Claude (median; 95th percentiles 215 and 139 ms).

Table 2. **Closed agents at a 5% review budget**, with monitors trained only on Gemma-4. Pooled AUROC and the share of violations caught when exactly 5% of the actions, the highest-scored, go to review (a cut set without labels). Best in each column in bold, second underlined.

	Claude Haiku 5.5		GPT-6-luna	
Monitor	AUROC	caught	AUROC	caught
Zero-shot 70B judge	0.570	0.0%	0.778	37.9%
Tool identity	0.857	42.4%	0.856	13.6%
TF-IDF (same labels)	<u>0.975</u>	<u>63.8%</u>	<u>0.964</u>	<u>49.0%</u>
Grey-box reader	**0.989**	**94.1%**	**0.996**	**98.5%**

4.6. Labels of a new policy transfer from open agents

A monitor learns the policies in its training data, so a new policy needs labels (Section A). These labels need not come from the monitored agent. Prompt injection on AgentDojo is a new policy for our monitor. Labels of it collected on two *open* agents transfer to the *closed* GPT-6-luna: AUROC 0.922 against the judge's 0.787 (+0.136 [0.078, 0.191]), and the share of injected calls caught doubles, from 28% to 58% at the label-free cut (36% to 92% at the benign-call cut of Figure 4b). In an exploratory check on the confirmation split, whose user tasks the training never saw, the result holds (0.929 vs 0.786, +0.143 [0.075, 0.227]).

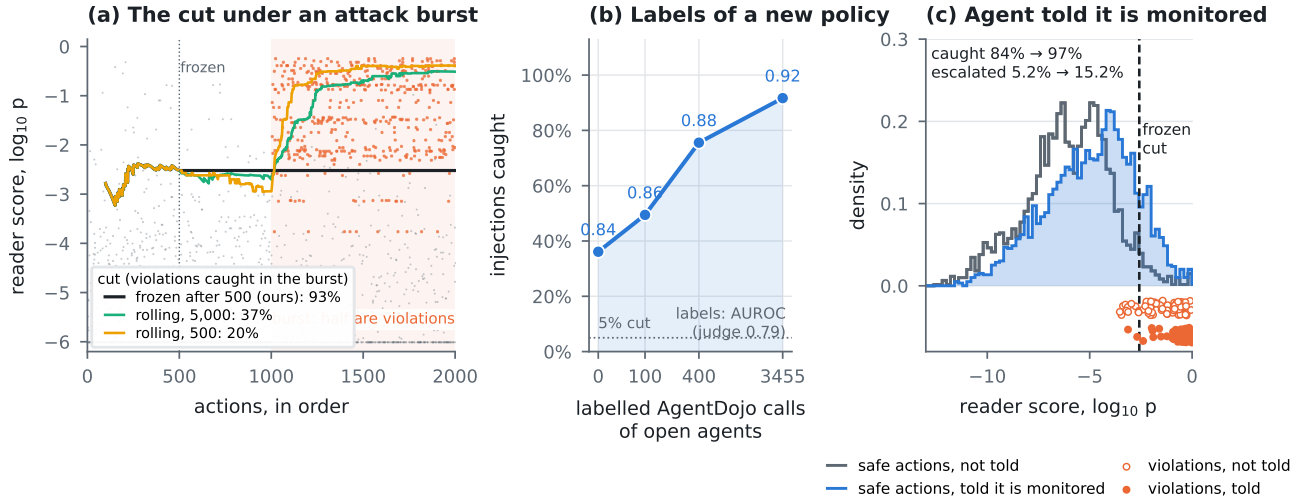


Figure 4. Robustness of the label-free cut, and transfer. (a) Claude Haiku 5.5's reader scores in order (grey; violations in orange): 1,000 normal actions, then a burst in which half the actions are violations (one run of the protocol behind Figure 5c). A cut frozen after 500 actions holds; cuts that keep moving rise with the burst and let it through (legend: violations caught in the burst; mean of 50 runs 94%, 36%, 20%). (b) Labels of prompt injection collected on two *open* agents, applied to the *closed* GPT-6-luna: injected calls caught at the 95th percentile of benign calls, against the number of labelled calls (numbers: AUROC). (c) Claude told, or not told, that it is monitored: the reader's scores of safe actions (histograms) and of violations (dots), and the cut frozen on Claude's earlier unlabelled traffic.

Table 3. **Live prevention** in the closed agents' tool loops (preregistered; each of 500, 450, 450 and 400 scenarios run without and with the gate). Episodes with an executed violation and episodes that reached the task goal, in %; Δ: gated − ungated violation rate (as a fraction) with its 95% interval (bootstrap over scenarios).

	Violations			Goals	
Agent, gate	none	gate	Δ [95% CI]	none	gate
GPT-6-luna, block	6.6	**0.2**	−.064 [−.088, −.044]	47.4	45.6
Claude, block	6.7	**0.2**	−.064 [−.087, −.044]	40.7	34.9
hold for review[a]	6.7	**0.0**	−.067 [−.091, −.044]	40.7	38.7
told, cut re-set[b]	6.5	**0.0**	−.065 [−.090, −.043]	38.5	35.5

[a]An oracle reviewer; same scenarios as the row above, run separately. [b]Block, with the cut set again on the told agent's unlabelled traffic: it escalated 4.7% [3.5, 6.1] of actions (budget 5%; the old cut 16.6%).

4.7. An agent that knows it is monitored

A deployed agent may know that a monitor reads its actions. We ran Claude Haiku 5.5 on 990 fresh scenarios twice, once with an oversight notice in its system prompt ("every tool call is reviewed by an automated safety monitor") and once without, with the shipped monitor and the cut frozen on Claude's earlier unlabelled traffic. Being told did not change how often Claude violated (−0.003 [−0.012, +0.005]) or reached its goal (37.6% vs 37.8%). The monitor caught 97.1% [92.9%, 100%] of the violations of the told agent and 83.6% of the others (+0.136 [0.050, 0.225]; both preregistered endpoints pass). The notice moved the whole score distribution up (Figure 4c): the frozen cut escalated 15.2% of the told agent's actions instead of 5.3%. Proposition 8 bounds this: the Kolmogorov distance between the told agent's scores and the warm-up scores is 0.24 (0.014 for the agent that was not told), and the escalation rate moved by 0.10. At each arm's own 5% budget the monitor catches 88.6% and 83.6%.

Set the cut again, and the budget is restored. A change of prompt is a shift, so Propositions 7 and 8 prescribe a new cut from the new, unlabelled traffic. We tested this live on 400 fresh scenarios, with the cut set from the told agent's earlier scores (no labels), then frozen. The new cut escalated 4.7% [3.5%, 6.1%] of the actions, inside the band fixed in advance around the predicted 5%, where the old cut would have escalated 16.6%, and the gate removed every executed violation (Table 3; both preregistered endpoints pass). Of the 54 blocked actions, 25 would have been violations.

An adaptive attacker defeats the monitor. An attacker who can query the score and rewrites only the free text of a violating call, keeping its target, lowered the share caught from 96% to 29%. In an exploratory diagnosis, in 81 of the 82 attacked messages the violation is the recipient, a list whose members the agent never checked; the attacker dropped the copied draft from two thirds of them, and every rewrite still violated. The monitor had caught these messages through that removable cue. Proposition 6 bounds every defense by the clean accuracy of the best monitor of the decisive fields; our canonical reader, which estimates it from below, catches 51–59% at 5%, because this family is absent from training. Two preregistered defenses reached 51% (the canonical reader) and 60% [52%, 68%] (training on rewrites) after fresh attacks, close to this estimate

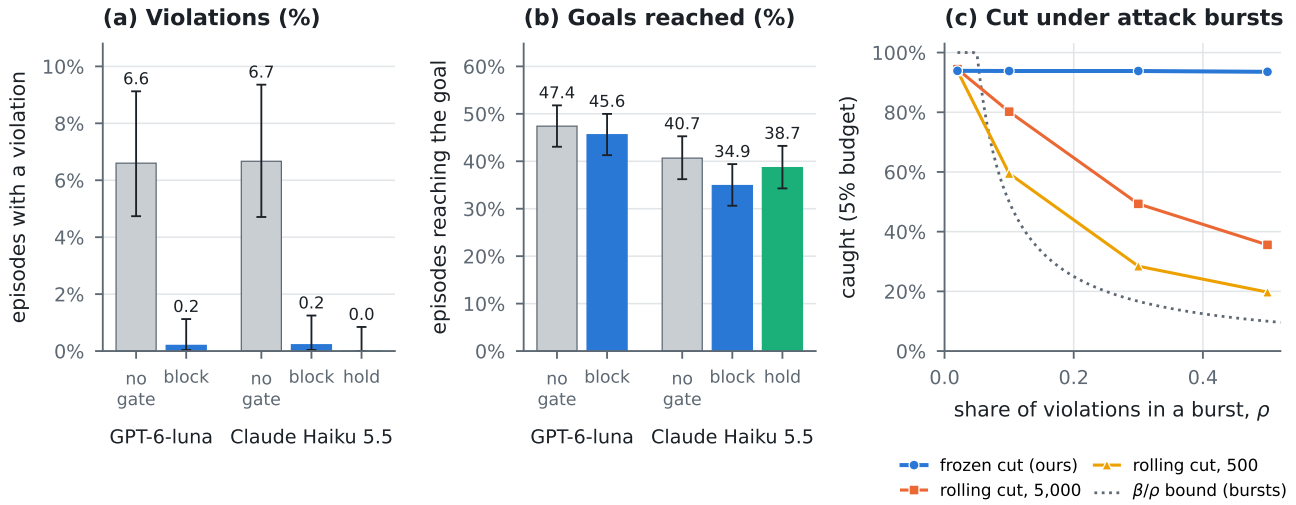


Figure 5. Prevention. (a) Episodes with an executed violation in the two closed agents' tool loops, without a gate, with blocking, and with holding for review (Claude); 95% Wilson intervals. (b) Episodes that reached the task goal in the same arms. (c) The share of violations caught at a 5% budget when a burst of attacks makes up a share ρ of the traffic: a cut frozen on earlier traffic (ours) against cuts that keep moving; dotted, the most a cut set on the burst itself can catch (Proposition 9).

(Section A). Robust monitoring needs labels that cover the decisive fields, and the score must stay hidden from the agent.

5. The tool

We release an SDK, a monitoring server and a website (links in the camera-ready version). A user imports the agent's logs (OpenAI or Anthropic format), labels actions, and trains a monitor in the browser; the logs never leave the machine. The trainer reports the AUROC with whole tasks held out, the share caught at the user's budget and a label-free cut; a review queue ranks unlabelled actions by risk (on 300 GPT-6-luna conversations, its first 25 were all violations; base rate 4.7%). A guard places the monitor in a tool loop, an MCP proxy or one of three agent frameworks. Python and the browser give the same vocabulary and near-identical scores. The text monitor takes 0.2 ms per action and the reader 41 ms on one GPU (Section D).

6. Related work

Probes on internals read truth, refusal and deception (Burns et al., 2023; Marks & Tegmark, 2023; Zou et al., 2023; Arditi et al., 2024; Goldowsky-Dill et al., 2025); CAPE probes an agent before each action (Fayyazi et al., 2026). We move the probe outside the agent. **Chain-of-thought monitors** catch reward hacking that action-only monitors miss (Baker et al., 2025; Korbak et al., 2025; Chen et al., 2025); Corollary 1 delimits when the reasoning adds nothing: for labels that the action and the visible facts decide. **AI control** audits a fixed budget of actions with a trusted monitor (Greenblatt et al., 2024); Propositions 7 and 9 give the budget a label-free, exact and burst-robust form. **Identifiability.** Latents made identifiable by supervised training are recovered linearly by complete encoders (Zhou & Klindt, 2026); Proposition 1 is the output-side counterpart. **Black-box access** recovers parts of a model from logits (Carlini et al., 2024; Finlayson et al., 2024), detects lies with follow-up questions (Pacchiardi et al., 2024; Cooney et al., 2026; Sam et al., 2025), forecasts behaviour from traces (Levy et al., 2026), and probes closed transcripts with open readers (conti et al., 2026). We add the theory, a direct test of replication against the true state, two closed agents, live prevention and preregistration.

7. Conclusion

A closed agent's hidden state cannot be read in its own coordinates, but an open model that reads the same tokens replicates 46–66% of its main components, as a conditional mean on which white-box probes score at least as well. Because the transcript and the action are sufficient for policy checks, a probe trained on an open agent predicts and prevents two closed agents' violations on its training benchmark, also when the agent knows it is monitored, and a cut set again on unlabelled traffic keeps the review budget at 5% after a shift. An attacker who queries the score still evades it; both preregistered defenses failed, and Proposition 6 bounds what any defense can reach. Limits and failed tests are in Section A.

Impact Statement

On the benchmark it was trained on, this work monitors closed AI agents before they act, from their transcripts alone; on a new benchmark it needs labels of the new policy. The intended effect is fewer harmful actions by deployed agents: in our tests, blocking the flagged actions cut the episodes with an executed violation from about 7% to 0.2% on two closed agents. We see four risks. (1) A monitor can give false confidence: it catches violations of the policies that it was trained on and can miss a new policy until that policy has labels (Section A); we state this in the paper and in the tool. (2) Blocking has a cost for users: goal completion fell by 1.8 points on GPT and 5.8 points on Claude. (3) An agent or attacker that can query the monitor's score can reword a violating call until it passes (Section 4.7); the score should stay hidden from the agent. (4) Methods that read a model's state from outside could help an attacker to probe a closed model; ours use only the transcripts that the operator already has and do not extract weights or private data. The code is released under the MIT license, and the browser tool does not upload the user's logs.

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A. Limitations and negative results

This appendix collects the limits of every claim and every test that failed or could not be done. The main text reports only results that met their preregistered criterion; the complete record, with each test's outcome as it was registered, is in Section H.

A.1. Scope of the claims

- **Replication is tested only where the true state is known** (four open agents). For closed agents we test prediction and prevention with a probe on the reader's state; the replica itself, mapped into Gemma-4's coordinates, scores lower on the closed agents (AUROC 0.959 and 0.976; exploratory). Replication is partial: 46–66% of the variance of the main components, 51–54% of each full layer on Gemma-4 and 30–46% on the other three agents.
- **The replica's gain is shrinkage, not the reader alone.** The true-state probe also scores higher on a TF-IDF replica than on the reader replica for the three new agents (within template 0.934, 0.977, 0.971 vs 0.924, 0.972, 0.969; exploratory): any conditional-mean estimate removes label-free variance.
- **The decisive information is largely in the tokens.** A probe on the true state also works on a TF-IDF replica (0.983), and TF-IDF with the same labels equals the reader on the training benchmark (within-template AUROC 0.997 vs 0.991 on Claude; pooled, 0.975 vs 0.989; the preregistered test that the reader beats it there failed). The reader's advantage is robustness to shift.
- **Most agent tests use one benchmark**, CAPE-Bench, whose tasks are synthetic. The headline closed-agent results (94% and 98% caught; live prevention) are on the benchmark the monitor was trained on.
- **Corollary 1 needs a label that the action and the visible facts decide.** It does not cover labels about later behaviour or beliefs (for later behaviour, the rollout forecast dominates every readout; Proposition 10).
- **Scores do not transfer between agents**: on Claude, a cut at probability 0.5 catches 3%. The cut must come from the monitored agent's own traffic (Proposition 7).
- **The live tests compare the gate with no gate**; no baseline gate ran in the loop. Offline, on the same agents' 991- and 1,000-episode runs, holding 5% of actions would have stopped 94%/98% (Claude/GPT) of the episodes with a violation with the reader, 64%/49% with TF-IDF, 0%/38% with the 70B judge and 42%/14% with tool identity (exploratory).
- **The monitor is not robust to an adaptive attacker.** An attacker who queries the score and rewrites free text evades it; both preregistered defenses failed (below). Keep the score hidden from the agent.
- **Blocking costs goals.** Goals reached fell from 47.4% to 45.6% (GPT) and from 40.7% to 34.9% (Claude); holding for an oracle reviewer gave 38.7%, and the preregistered test that review costs at most 3 points was not shown (-0.020 [-0.051, $+0.007$]). A human reviewer is slower and makes errors.

A.2. Tests that failed or could not be done

Test	Outcome
Interrogation state alone vs a white-box probe	The agent's answers alone do not match a probe on its internals at the start of the turn (failed).
Resampling scored by a deployable judge	A 70B judge in place of the environment checker: 0.782, below the probe (0.846); the label function is the bottleneck (failed).
Claude, actions after a reveal	0 violations in the preregistered units: Claude violates at or without the reveal (not tested; the all-actions test replaced it).
Reader vs TF-IDF in distribution	TF-IDF equals the reader on the training benchmark (failed; it is worse under shift).
A policy family removed from training	The ranking holds (0.912 vs 0.570), but 0 of 26 violations are escalated at a 5% budget without labels of the policy.
Labels for a new policy, prospective	The held-out credentials policy needed no labels (18 of 18 caught with 0), so there was nothing to fix (failed as registered).
AgentDojo with spotlighting	Attacks succeeded in 2.1% of episodes (4 injected calls; not tested).
AgentDojo without a defense, closed agent, no AgentDojo labels	0.840 vs judge 0.787, $+0.052$ [-0.026, $+0.130$] (failed); with labels of open agents the same test passes (Section 4.6).
Hold for review: goal cost	Not shown within 3 points (above).

Test	Outcome
Browser-size reader vs the 8B reader	-0.097 [-0.115, -0.079] AUROC on the closed agents' CAPE-Bench actions, against a margin of -0.05 (failed); offered for new kinds of tasks only.
A second outside benchmark, τ^2-bench (an airline policy)	
Claude alone	9 violations, below the preregistered 15 (not tested).
Two closed agents, a persistent customer	TF-IDF 0.987 within tool, the reader 0.738 (-0.250 [-0.518, -0.077]; failed). Every test task also occurred in training.
The same, test tasks removed from training	Neither monitor transfers: reader 0.535, TF-IDF 0.519 (both failed); TF-IDF with entities masked 0.366. The 0.987 was memorisation of the tasks. A τ^2 violation breaks a written rule over facts of the case (a booking older than 24 hours, the cabin), which a linear readout of other tasks' labels does not learn.
A zero-shot reasoning judge (Qwen3-32B)	0.804 within tool on the same writes, but 7 violating tasks; both endpoints failed (wide intervals).
The same judge, four frontier agents	894 writes: it separates violating from allowed writes of the same tool (0.766, passes), but not significantly better than the reader trained on other tasks (0.646; $+0.120$ [-0.047, $+0.274$], failed); at a 5% budget the reader caught 18.4%, the judge 11.7%, tool identity 13.0% (exploratory).
Robustness	
An adaptive attacker who can query the score	Rewriting only the free text of 134 real closed-agent violations (two rounds of 16, score feedback; every rewrite still violates) cut the shipped reader's catch rate from 96.3% to 29.1% [21.6%, 36.6%] (failed), and the reader was not more robust than TF-IDF (-0.052 [-0.149, $+0.037$]). The rewrites keep the decisive field and add self-reported approvals. The two fixes below were tested against fresh attacks.
A monitor that reads the canonical action	Free-text fields replaced by a placeholder and quoted prose removed, so that rewording cannot move the score (Corollary 1, canonical form). The invariance held: on messages the score did not move under attack. But clean catch at 5% fell from 94.1% to 51.5% (Claude) and from 98.5% to 59.1% (GPT); pooled -0.410 [-0.493, -0.311], AgentDojo -0.029 [-0.080, $+0.010$], and 50.8% [42.5%, 59.0%] were caught after the attack (all three endpoints failed). In 81 of the 82 attacked messages the violation is the recipient (a list whose members the agent never checked), and the family is absent from training: the shipped monitor had caught these messages through a copied draft in the body, a cue that the canonical form removes and the attacker drops (Proposition 6). It is an option in the tool, with this cost, not the default.
Training against the attack	Rewrites of both violating and safe training actions (4,554, labelled by the checker) added to training, then a fresh attack that is told the defense: 59.7% [51.5%, 67.9%] caught after the attack (failed; 92.5% before). The shipped monitor caught 57.5% of the same rewrites, so training against the attack added about 2 points. Clean catch -0.037 [-0.078, $+0.016$] (failed) and AgentDojo $+0.012$ [-0.050, $+0.071$] (failed at the margin). Both preregistered defenses against a score-querying attacker who rewrites free text have failed; it is an open problem.
The interrogation state in chat	
Forecasting Claude's refusals	Ties a graded text moderator in a prospective test (0.873 vs 0.855); it adds precision only in combination (Section G).

A.3. Design order

Several tests were designed after earlier results: the second τ^2-bench tests after the first, the reasoning judge after the unseen-task test, the canonical form after the adaptive attack, and adversarial training after the canonical form. Each was preregistered and locked before its own data existed, and the later locks were published first, but their hypotheses were not independent of what we had seen.

B. Proofs

Notation is that of Section 2. $\sigma(X)$ is the σ-field generated by X. A score is *strictly proper* if its expectation is uniquely maximised by the true conditional probability.

Proposition 10 (behavioural sufficiency)**.** *Let $Y = g(\tau)$ be defined by the future τ that the agent and the environment generate from context x, and let $\phi(x)$ carry no information about the future sampling noise given x. Then $\pi^*(x) = P(Y = 1 \mid x)$ has a maximal ROC curve and a minimal risk under every strictly proper score among all forecasts $f(\phi(x))$.*

Proof. By the assumption and the tower property, $\mathbb{E}[Y \mid \phi(X)] = \mathbb{E}[\pi^*(X) \mid \phi(X)]$. Under a strictly proper score the Bayes act given $\sigma(X)$ is $\pi^*(X)$, and $\sigma(\phi(X)) \subseteq \sigma(X)$; less information cannot raise the expected score. For the ROC curve, by the Neyman–Pearson lemma the most powerful test of each size thresholds the likelihood ratio $p(x \mid Y{=}1)/p(x \mid Y{=}0)$, a strictly increasing function of $\pi^*(x)$; every test based on $f(\phi(X))$ is a test based on X. The same holds within any stratum that is a function of X, hence for the within-stratum AUROC. □

Proposition 1 (gauge). Write a pre-norm residual network as $h^{(\ell+1)} = h^{(\ell)} + W_{\text{out}}^{(\ell)} F^{(\ell)}\big(W_{\text{in}}^{(\ell)}\, N(h^{(\ell)})\big) + b_{\text{out}}^{(\ell)}$, with embedding $h^{(0)} = E(x)$, output logits $U\,N(h^{(L)})$ and $N(h) = h/\mathrm{rms}(h)$ after folding the normalisation gains into W_{in} and U. For orthogonal G, $\mathrm{rms}(Gh) = \mathrm{rms}(h)$, so $N(Gh) = G\,N(h)$. Replace $E \mapsto GE$, $W_{\text{out}} \mapsto GW_{\text{out}}$, $b_{\text{out}} \mapsto Gb_{\text{out}}$, $W_{\text{in}} \mapsto W_{\text{in}}G^\top$ and $U \mapsto UG^\top$. By induction on ℓ the new residual stream is $Gh^{(\ell)}$, since $W_{\text{in}}G^\top N(Gh) = W_{\text{in}}N(h)$ and the new update is G times the old one; the logits $UG^\top N(Gh^{(L)}) = UN(h^{(L)})$ are unchanged, so the output distribution is unchanged on every input. At a layer read only through linear maps (no normalisation in between), the same argument holds for every invertible G with G^{-1} in place of $G^\top$. Any estimator of hidden coordinates that uses only outputs takes the same value on both networks, so it cannot distinguish h from Gh; a probe $w^\top h$ on one network equals $(G^{-\top}w)^\top(Gh)$ on the other. *Other normalisations.* LayerNorm subtracts the mean, $N(h) = Ph/\mathrm{rms}(Ph)$ with $P = I - \frac{1}{d}\mathbf{1}\mathbf{1}^\top$; $N(Gh) = GN(h)$ holds for orthogonal G with $G\mathbf{1} = \mathbf{1}$, which still form a group of dimension $(d-1)(d-2)/2$. A normalisation *after* a block, $\mathrm{diag}(\gamma)N(\cdot)$ added to the stream, needs $G\,\mathrm{diag}(\gamma) = \mathrm{diag}(\gamma)\,G$: G must commute with the gains, so for distinct gains only sign flips remain, and for equal gains any orthogonal map within the block of equal gains. From exact full-vocabulary log probabilities the final hidden state is identified only up to such a G (Carlini et al., 2024). □

Proposition 2 (Fisher–Rao coordinates). The Fisher information of a Bernoulli(p) observation is $1/\big(p(1-p)\big)$. With $\vartheta = 2\arcsin\sqrt{p}$, $\mathrm{d}\vartheta/\mathrm{d}p = 1/\sqrt{p(1-p)}$, so $\mathrm{d}\vartheta^2 = \mathrm{d}p^2/\big(p(1-p)\big)$: the map is an isometry onto an interval of the Euclidean line, and independent coordinates give the product metric on $[0,\pi]^m$. For $\hat{p}$ the mean of n answers, the delta method gives $\mathrm{Var}(\hat{\vartheta}) \approx (\mathrm{d}\vartheta/\mathrm{d}p)^2\, p(1-p)/n = 1/n$; the first-order expansion needs $np(1-p) \gg 1$ (near $p \in \{0,1\}$ the variance is smaller). □

Proposition 3 (battery design). Each chosen question splits at least one class, so the number of classes grows by at least one per question and starts at one; the loop stops when no question in the pool splits a class. At that point two contexts in the same class agree on every pool question (otherwise that question would split their class), and two contexts in different classes disagree on a chosen question; so the final partition is the one that the whole pool induces, reached with at most (number of its classes $-\,1$) questions. If the answers come from a finite automaton with minimal automaton M, two prefixes give the same answer to every suffix iff they reach the same state of M (Myhill–Nerode); when the pool holds a distinguishing suffix for every pair of states, the pool partition is the partition by minimal state. States that never change an output are behaviourally inert, so no black-box method identifies more. On a program compiled into a transformer (a bracket-balance automaton: 21 reachable states plus a dead state), the algorithm chose 21 questions from a pool of 2,101 and separates all 22 classes. □

Proposition 4 (access and replica). (a) $\phi_t = \Phi_\theta(Z_t)$ is $\sigma(Z_t)$-measurable, so $\sigma(Z_t, \phi_t) = \sigma(Z_t)$ (for every label), and Proposition 10 applied with $X = Z_t$ gives the first claim. Under log loss the Bayes risk given a σ-field $\mathcal{G}$ is the conditional entropy $H(Y \mid \mathcal{G})$, and $H(Y \mid V_t) - H(Y \mid V_t, H_t) = I(Y; H_t \mid V_t)$. (b) By the law of total covariance, $\mathrm{Cov}(\phi) = \mathbb{E}\,\mathrm{Cov}(\phi \mid \psi) + \mathrm{Cov}\,\big(\mathbb{E}[\phi \mid \psi]\big)$, and the first term is positive semidefinite. For component k, $\mathrm{Var}(\phi_k) = \mathbb{E}(\phi_k - \hat{\phi}_k)^2 + \mathrm{Var}(\hat{\phi}_k)$, so $\mathrm{Var}(\hat{\phi}_k) = R_k^2\,\mathrm{Var}(\phi_k)$. For the best linear predictor $\hat{\phi}_k = a_k^\top\psi + b_k$ the residual is uncorrelated with $\hat{\phi}_k$ (the normal equations), and the same decomposition holds in sample. On fresh steps neither identity is exact: Figure 3d shows the ratio closely tracking R_k^2 on 128 components (correlation 0.99, slope 0.89), above R_k^2 on 80 of them. (c) Conditional expectation is linear: $\mathbb{E}[w^\top\phi \mid \psi] = w^\top\mathbb{E}[\phi \mid \psi]$. □

Proposition 5 (the replica Rao–Blackwellises a probe). $s = w^\top\phi$ is a function of ϕ, so $Y \perp\!\!\!\perp \phi \mid \psi$ gives $Y \perp\!\!\!\perp s \mid \psi$ and hence $\mathbb{E}[s \mid \psi, Y] = \mathbb{E}[s \mid \psi] = \hat{s}$. For ℓ convex in its first argument, conditional Jensen gives $\mathbb{E}[\ell(s, Y) \mid \psi, Y] \geq$

$\ell(\mathbb{E}[s \mid \psi, Y], Y) = \ell(\hat{s}, Y)$; take expectations. For a pair of independent draws $(\phi_i, \psi_i, Y_i = 1)$ and $(\phi_j, \psi_j, Y_j = 0)$, $\mathbb{E}[s_i - s_j \mid \psi_i, \psi_j, Y_i, Y_j] = \hat{s}_i - \hat{s}_j$ by the same conditional independence and the independence of the draws, and Jensen applies to $\ell(s_i - s_j)$. The AUROC is the expectation of the non-convex step $\mathbf{1}\{s_i > s_j\}$, so the result covers its convex surrogates (logistic, exponential, hinge), not the AUROC itself. □

Corollary 1 (sufficiency of the proposed action). Let $A_t = a(Z_t^-, \xi)$ with $\xi \perp\!\!\!\perp (Z_t^-, W)$, and let V_t be a function of Z_t^-.

1. $\xi \perp\!\!\!\perp (Z_t^-, W)$ gives $\xi \perp\!\!\!\perp W \mid Z_t^-$, hence $\xi \perp\!\!\!\perp W \mid (Z_t^-, V_t)$; with $W \perp\!\!\!\perp Z_t^- \mid V_t$, contraction gives $W \perp\!\!\!\perp (Z_t^-, \xi) \mid V_t$.
2. A_t is a function of (Z_t^-, ξ), so $W \perp\!\!\!\perp (Z_t^-, A_t) \mid V_t$, and by weak union $W \perp\!\!\!\perp Z_t^- \mid (V_t, A_t)$.
3. $Y = g(A_t, V_t, W)$, so $Y \perp\!\!\!\perp Z_t^- \mid (V_t, A_t)$: $P(Y \mid Z_t^-, A_t) = P(Y \mid V_t, A_t)$.
4. A monitor that reads internals at or before the action, and possibly the action, sees a σ-field inside $\sigma(Z_t^-, A_t)$. Under a strictly proper score the Bayes risk given a coarser σ-field is no smaller, and by step 3 the Bayes act given $\sigma(Z_t^-, A_t)$ is $P(Y \mid V_t, A_t)$; so its Bayes risk is at least that of the best monitor of (V_t, A_t). □

The inequality can be strict: if A_t is random given Z_t^- and Y depends on which action is sampled, a probe read before the action must average over the actions. *Canonical form.* If moreover $Y = g(\kappa(A_t), V_t, W)$ for a function κ of the action, the same steps with $\kappa(A_t)$ in place of A_t show that $(V_t, \kappa(A_t))$ is sufficient, and every monitor of $(V_t, \kappa(A_t))$ is invariant to any change of A_t that keeps $\kappa(A_t)$. The premise is about the label. In our attacked family it held: the checker labels a send by its recipient. The canonical monitor failed there because that field was never learned: the family is absent from training (Section A).

Proposition 6 (a ceiling under rewording). Write an action as (κ, t), with t its free text, and let $T_0(V, \kappa)$ be the set of texts that safe actions with context V and decisive part κ use. Define $m(V, \kappa) = \inf_{t \in T_0(V,\kappa)} s(V, \kappa, t)$, and $m = +\infty$ where T_0 is empty. On a safe action $t \in T_0(V, \kappa)$, so $m \le s$ and $m \ge c$ implies $s \ge c$: m escalates no more safe actions than s. Under the attack, a violation with (V, κ) is rewritten to the text in $T_0(V, \kappa)$ with the lowest score (allowed by the assumption; take a sequence if the infimum is not attained), so its attacked score is $m(V, \kappa)$, and it is escalated iff $m(V, \kappa) \ge c$. As m depends only on (V, κ), the attack does not change it, and its share of violations caught on clean traffic equals the share that s catches under the attack. Where T_0 is empty the attacker cannot imitate a safe action, and $m = +\infty$ escalates. □

Proposition 7 (a frozen cut). For exchangeable scores without ties, the rank of S_{n+1} among $S_1, \ldots, S_{n+1}$ is uniform on $\{1, \ldots, n+1\}$. $S_{n+1} > S_{(k)}$ holds iff its rank exceeds k, which has probability $(n+1-k)/(n+1)$; with $k = \lceil (1-\beta)(n+1) \rceil$ this lies in $(\beta - \frac{1}{n+1}, \beta]$. For i.i.d. scores with a continuous distribution function F, a new score exceeds $\hat{c}$ with probability $1 - F(\hat{c})$ given the warm-up; $F(S_{(k)})$ is the k-th order statistic of n uniforms, $\mathrm{Beta}(k, n+1-k)$, so over warm-up draws $1 - F(\hat{c}) \sim \mathrm{Beta}(n+1-k, k)$, with mean $(n+1-k)/(n+1)$ and variance $k(n+1-k)/\big((n+1)^2(n+2)\big)$; for $n = 500$, $\beta = 0.05$: $k = 476$, mean 0.0499, sd 0.0097. The shipped cut is the linearly interpolated empirical quantile at $1 - \beta$, which lies in $[S_{(m)}, S_{(m+1)}]$ with $m = \lfloor 1 + (n-1)(1-\beta) \rfloor$. Its escalation probability is therefore in $\left[\frac{n-m}{n+1}, \frac{n+1-m}{n+1}\right]$, and $(n-1)(1-\beta) \le m \le 1 + (n-1)(1-\beta)$ puts it in $\left[\beta - \frac{2\beta}{n+1}, \beta + \frac{2(1-\beta)}{n+1}\right]$, within $\frac{2}{n+1}$ of β (for $n = 500$: between 0.0499 and 0.0519). □

Proposition 8 (shift). For every c, $Q(S > c) - P(S > c) = F_P(c) - F_Q(c)$, whose absolute value is at most $K(P, Q)$. Apply it at $c = \hat{c}$ and combine with Proposition 7; apply it to the laws of the violations' scores for the share caught. □

Proposition 9 (bursts). A cut at the top β of a window of N actions escalates $\lceil \beta N \rceil$ of them; the window holds ρN violations, so at most $\beta N/(\rho N) = \beta/\rho$ of them are escalated (up to rounding). A frozen cut $\hat{c}$ escalates a violation with probability $P(S \ge \hat{c} \mid Y{=}1)$, the monitor's true-positive rate at $\hat{c}$, which does not depend on how many violations the traffic holds when the score laws do not depend on ρ. A rolling window that holds only part of a burst has a smaller effective share than the burst, so it can catch more than β/ρ of the burst's violations; the bound is on the window. □

C. Experimental details

Agents. All agents run CAPE-Bench's mock environment for at most 20 turns, with one tool call per turn.

- Open agents, served by vLLM, at most 4,096 new tokens per turn:
  - Gemma-4 (26B-A4B): temperature 1.0, top-p 0.95, top-k 64, thinking on.
  - gpt-oss-20b: temperature 1.0, top-p 1.0, reasoning effort "medium".

  The episodes of the three additional open agents come from the released CAPE-Bench rollouts.
- Closed agents use their public tool-use APIs:

Algorithm 2 Battery design from black-box answers

Require: contexts $x_1..x_N$; pool of questions $\mathcal{Q}$; answers $a(x, q) \in \{0, 1\}$ from the model's outputs only
1: $C \leftarrow$ one class holding all contexts; $Q \leftarrow ()$
2: **loop**
3: $q^\star \leftarrow \arg\max_{q \in \mathcal{Q}}$ #classes of C refined by $a(\cdot, q)$
4: **if** $q^\star$ splits no class **then return** Q {Proposition 3}
5: append $q^\star$ to Q; $C \leftarrow C$ refined by $a(\cdot, q^\star)$
6: **end loop**

- Claude Haiku 5.5: thinking disabled, effort "low", at most 1,024 tokens per turn.
- GPT-6-luna: reasoning effort "none", no parallel tool calls, at most 2,048 tokens per turn.

The shared spend ledgers record $24.78 of API use in total (Anthropic $19.86, OpenAI $4.92), including tests of the released tool and a 40-episode pilot ($0.10) of Claude on AgentDojo that was not analysed.

Units. A unit is one proposed action that is not a read, after the step that reveals the deciding fact. Each episode contributes its actions up to and including its first violation. All violations are kept. Safe actions are subsampled at random with seed 0, so selection depends only on the past.

Reader input. The reader prompt holds four parts:

- the operating instructions;
- the user task;
- each earlier call with its result (results cut to 2,000 characters);
- the proposed action.

A context of 30,000 characters or more keeps its first 6,000 and last 24,000 characters. The prompt goes through Qwen3-8B's chat template with thinking off and is cut to its last 12,000 tokens. The feature is the residual stream at the last prompt token, at hidden-state index 32 (the output of block 31 of 36), in bf16.

Monitor. The monitor standardises the features, keeps the top 256 principal components, and fits a logistic regression with $C = 0.05$. It is trained on Gemma-4's CAPE-Bench steps only (17,623; development and fresh). The review cut has no labels. Offline (Table 2), it sends exactly 5% of the agent's actions in the test set, the highest-scored, to review (actions tied at the cut count in proportion). Live, it was the 95th percentile of the agent's earlier unlabelled actions, then frozen; the tool freezes it after the first 500. A rolling cut fails under bursts of similar actions.

Replica.

- **Target:** the agent's residual stream at its end-of-reasoning token, at 5 depths (block $\mathrm{round}(nf) - 1$, $f \in \{0.5, 0.62, 0.72, 0.82, 0.9\}$). It is standardised and reduced to 128 principal components, all fit on development.
- **Map:** ridge regression, with $\alpha \in \{1, 10, 10^2, 10^3, 10^4\}$ chosen by 5-fold cross-validation grouped by scenario.
- **Probe on the true state:** standardise, then logistic regression with $C = 0.05$, on the 128 true components.

Text baseline. TF-IDF of the context and of the action, kept as two separate blocks:

- word 1- and 2-grams, minimum document frequency 3, at most 200,000 features each, sublinear term frequency;
- logistic regression with $C \in \{0.1, 1, 10\}$, chosen by 5-fold cross-validation grouped by scenario.

The browser tool runs the same model. On 302 real agent steps it gives the same text as Python, and its held-out AUROC is 0.9887 (Python 0.9886).

Statistics.

- AUROC is computed within strata (template and tool, or suite) and averaged with weights by the number of positive–negative pairs.
- Intervals are percentile intervals from 1,000 bootstrap resamples of scenarios or episodes. The seeds are fixed in each preregistration.
- Each preregistration (the document and its analysis code) is locked by its SHA-256 hash before its data exist. One script checks every lock and every number in this paper again from the saved results.

Compute. All GPU work runs on NVIDIA H100 NVL GPUs (96 GB), one job at a time except for one 8-hour period

with two (and one accidental 4-second overlap): PyTorch 2.11, Transformers 5.14.1 (5.20.1 for Muse-Glimmer, whose architecture the earlier version does not know), scikit-learn 1.9.0. The reader takes 41 ms for one action on one GPU; the median server round trip in the three blocking live tests was 89, 80 and 83 ms.

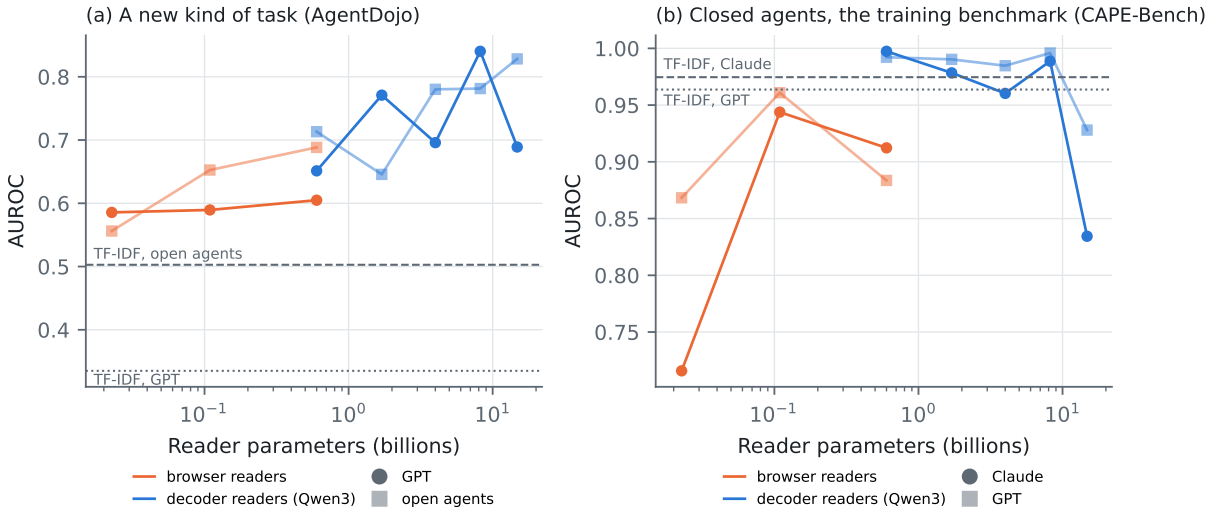
D. The released tool



Figure 6. The released tool. (a) Labelling imported agent logs in the browser, with the review queue: the monitor trained on the labels so far puts the riskiest unlabelled action first. (b) The replay of Claude Haiku 5.5: each of its 3,128 actions in the monitor's space, with the label-free cut as a plane.

E. Smaller readers

All readers are trained on the same 17,623 Gemma-4 CAPE-Bench steps (1,339 violations); none is trained on a test set. Browser encoders: all-MiniLM-L6-v2, all-mpnet-base-v2 (three 1,000-character views) and Qwen3-Embedding-0.6B (one instruction-prefixed text, last-token pooling), each with a standardised logistic probe. Decoder readers: Qwen3-0.6B, 1.7B, 4B and 14B at the shipped reader's relative depth, with the shipped probe. The selected browser encoder, Qwen3-Embedding-0.6B, beats TF-IDF on the closed agent's AgentDojo calls (0.605 vs 0.335; +0.270 [0.134, 0.451]), so the website offers it for new kinds of tasks; on CAPE-Bench it is 0.097 below the 8B reader (Section A). In the browser it runs as a float16 ONNX model whose features match PyTorch's (cosine 1.000 on 40 steps); the 8-bit model's do not (0.852).



F. The probe on the replica

The numbers behind Proposition 5 and Section 4.2, for the four agents of the replication tests.

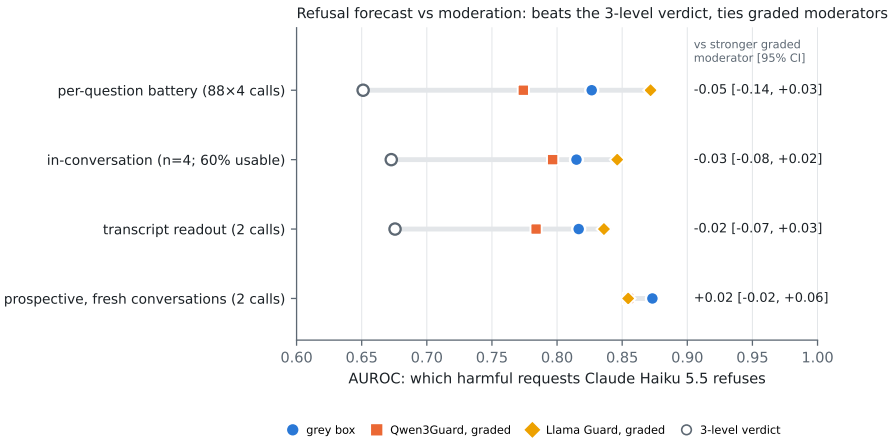
Table 5. A probe trained on the agent's *true* states, scored on the true states and on the reader's replica (rep.; fresh steps; exploratory). AUROC pooled and within template; the better of each pair in blue.

	pair risk ↓		AUROC		in template	
agent	true	rep.	true	rep.	true	rep.
Gemma-4	.018	**.015**	.994	**.996**	.984	**.987**
gpt-oss	.078	**.045**	.972	**.986**	.900	**.924**
Qwen3.8	.044	**.035**	.986	**.990**	.955	**.972**
Muse	.056	**.022**	.982	**.996**	.929	**.969**

Pairwise risk: $\mathbb{E}\log(1+e^{-(s_i-s_j)})$ over violating i and safe j.

G. The interrogation state in chat

On Claude Haiku 5.5, readouts of the interrogation state trained only on open models forecast which harmful requests Claude refuses. Against a three-level moderation verdict the forecast wins (three preregistered tests); against graded moderators it ties (prospective, 923 conversations: 0.873 vs 0.855). Combined with a moderator, it doubles the precision of an alarm for harmful requests that the model answers (0.535 vs 0.274).



H. The preregistration record

Every confirmatory test, as in the released record (generated from it by a script; file paths and commit ids are replaced for review). Locks (SHA-256 of the preregistration and its code) and outcomes are in the released record; lock times are self-reported. Endpoints are numbered (i), (ii), ... in the order of each preregistration. Later tests changed two readings fixed in advance. First, the unseen-task test showed that labels do not transfer on tau2-bench through word counts: the test tasks had been in training. Second, "checks rules" in the frontier-agent judge test means only that the judge separates violating from allowed writes of the same tool.

Test	Outcome (verbatim from the record)
Interrogation state, confirmation open LLMs ((i)–(iii))	**PASS all, after deviation 1**: (i) +0.120 [0.115, 0.126]; (ii) +0.057 [0.045, 0.073] (7/8 models); (iii) +0.359 [0.310, 0.415]; Holm p < 0.0005 each. First run: (i) valid PASS, (ii)/(iii) invalid (numerical; see note) ([released file])
Resampling vs a probe agents ((i)–(iii); addendum 1: (iv)–(v))	**Claim (a) CONFIRMED** ((i), (ii) pass: resampling (16 rollouts) − white-box probe +0.063 [0.033, 0.094]); claim (b) NOT confirmed ((iii) fails: Ψ − white-box probe −0.021 [−0.054, 0.009]); addendum **(iv) passes** ((iv)+Ψ − white-box probe +0.049 [0.020, 0.080]); (v) fails ((ii)+Ψ − resampling (16 rollouts) −0.023 [−0.053, 0.006]) ([released file])

Test	Outcome (verbatim from the record)
Interrogation state, open chat models Othello recovery ((i)–(iii))	**PASS all**: (i) R^2 0.967 [0.967, 0.968] > 0.90; (ii) ΔR^2 +0.094 [0.086, 0.105], Holm p < 0.0005; (iii) Δ occupied acc +0.043 [0.040, 0.046], Holm p < 0.0005. White-box reference 0.983 ([released file])
Claude chat readouts Claude Haiku 5.5 readout transfer ((i)–(iii))	**all pass**: (i) 0.993 [0.983, 0.999]; (ii) 0.997 [0.990, 1.000]; (iii) 0.935 [0.868, 0.982] ([released file])
Claude refusals vs moderation Claude Haiku 5.5: grey box vs text moderation ((i)–(iii))	**(i) pass** 0.923 [0.884, 0.957]; **(ii) (key) pass** grey box − moderation +0.176 [0.060, 0.287] (0.827 vs 0.651); (iii) not tested (1/40 over-refusals) ([released file])
Claude, one-call grey box Claude Haiku 5.5: one-call grey box + compliance alarm ((i)–(iii))	**all pass** vs the preregistered 3-level verdict: (i) +0.142 [0.081, 0.203] (0.815 vs 0.673); (ii) AP +0.230 [0.161, 0.307]; (iii) AP +0.081 [0.036, 0.140]. Against GRADED moderators (i) ties (Qwen3Guard +0.018 [−0.043, +0.081]; Llama-Guard −0.031 [−0.082, +0.020]); the alarm gain holds ([released file], [released file])
Claude, transcript readout Claude Haiku 5.5: transcript-framing grey box ((i)–(ii))	**both pass**: (i) grey box − moderation on harmful refusals +0.141 [0.082, 0.199] (0.817 vs 0.676); (ii) compliance-risk AP (Claude-moderator) +0.115 [0.076, 0.179] ([released file])
When the internals are read time: internals at CAPE's decision point vs black box ((i)–(iii))	**all pass**: (i) internals at the decision − resampling at the start of the turn +0.075 [0.044, 0.106]; (ii) resampling at the decision − internals at the decision +0.021 [0.011, 0.035]; (iii) internals at the decision − white-box probe at the start of the turn +0.138 [0.107, 0.163] ([released file])
Claude vs graded moderators Claude Haiku 5.5: the Live tool's alarm, prospective ((i)–(iii))	**(i) pass** (alarm precision 0.535 vs 0.274 for the moderator alone, +0.261 [0.215, 0.311]); **(ii) FAIL** (grey box 0.873 vs graded Qwen3Guard 0.855, +0.018 [−0.023, 0.062]); **(iii) FAIL** (multi-turn 0.894 vs 0.906) ([released file])
Resampling with a deployable judge deployable monitors: LLM judge instead of the oracle checker ((i)–(iii))	**(i) FAIL** (judge-scored resampling 0.782 < turn-start probe 0.846, −0.065 [−0.115, −0.019]); **(ii): probe better** (CAPE probe 0.976 > judge on actual action 0.910, −0.067 [−0.104, −0.036] for the judge); **(iii) FAIL** (resampling hurts the action judge, −0.049) ([released file])
Reader vs the agent's internals reader trained on labels vs the agent's internals ((i)–(ii))	**both pass**: (i) reader − internals at the decision +0.015 [0.007, 0.026] (0.994 vs 0.979); (ii) reader − judge +0.085 [0.052, 0.127] ([released file])
Unseen task families agent internals vs transcript reader on unseen scenario families ((i)–(ii))	**(i) pass** (reader − agent internals on unseen families −0.007 [−0.014, −0.0001], margin −0.02); **(ii): agent probe better than zero-shot judge** (+0.062 [0.028, 0.101]) ([released file])
Another open agent reader trained on Gemma, monitoring gpt-oss ((i))	**(i) pass**: Gemma-trained reader on gpt-oss 0.915 vs zero-shot judge 0.833, +0.082 [0.043, 0.129] ([released file])
AgentDojo, no labels reader monitor on AgentDojo prompt injection, zero transfer training ((i)–(ii))	**(i) pass**: reader 0.781 vs zero-shot judge 0.537, +0.249 [0.136, 0.369]; **(ii) reader better than CAPE's gate**: gpt-oss calls as the prereg text specifies, 0.823 vs 0.601, +0.222 [0.143, 0.303] (locked code's broader subset: +0.264 [0.174, 0.359]) ([released file])
Claude, after a reveal Claude Haiku 5.5 as an agent, monitored by a reader trained only on Gemma ((i))	**(i) NOT TESTED**: 0 unsafe units under the preregistered post-reveal rule (Claude's 68 violations all happen at or without the reveal); see "Closed agent: Claude"
Closed agent: Claude Claude as an agent, all non-read actions ((i'))	**(i') pass**: Gemma-trained reader on Claude 0.991 vs zero-shot judge 0.633, +0.360 [0.302, 0.413]; 95.6% of violations caught at 5% escalation with a cut set on labelled safe actions (judge 0%); at the label-free cut used elsewhere, 94.1% ([released file], [released file])
Reader vs TF-IDF trained TF-IDF monitor on the reader's own input, same labels ((i) Claude, (ii) unseen families)	**(ii) pass, (i) FAIL**: unseen families reader 0.969 vs TF-IDF 0.950 (+0.019 [0.007, 0.032]); Claude reader 0.991 vs TF-IDF 0.997 (−0.006 [−0.027, +0.007]). AgentDojo (reported) 0.781 vs 0.503 ([released file])
A policy removed from training Claude on task families the monitor never saw ((i) pooled, (ii) within)	**(i), (ii) pass** (vs zero-shot judge): pooled 0.912 vs 0.570 (+0.342 [0.311, 0.371]); within 0.948 vs 0.633 (+0.318 [0.265, 0.374]). But at the label-free cut the unseen restart family is caught 0/26 ([released file])
Closed agent: GPT-6-luna second closed family: gpt-6-luna as agent, frozen monitors ((i) pooled vs judge, (ii) vs tool identity, (iii) within tool)	**(i), (ii), (iii) all pass**: GPT-6-luna, 3,627 actions, 66 unsafe in 6 families. Reader pooled 0.996 vs judge 0.778 (+0.217 [0.174, 0.265]); vs tool identity +0.141 [0.111, 0.171]; within tool vs judge +0.039 [0.015, 0.066]. 98.5% caught at the label-free 5% cut (judge 37.9%) ([released file])

Test	Outcome (verbatim from the record)
Labels for a new policy labels needed for a new policy, prospective on GPT ((i) 40 vs 0 labels, (ii) catch at 40 $\geq$ 0.5)	**(i) FAILS, (ii) passes**: with 0 labels of the held-out credentials policy, the reader already caught 18/18 of GPT's violations of it, so added labels had nothing to fix (40 − 0: 0.0). The "A policy removed from training" curve does not replicate on this family ([released file])
AgentDojo with a defense a closed agent on an external benchmark: GPT-6-luna on AgentDojo, frozen monitors ((i) reader vs judge)	**(i) NOT TESTED**: GPT-6-luna resisted the injections (attack success 2.1%), leaving 4 injected calls (minimum 10). Utility on benign tasks 0.862 ([released file])
AgentDojo, closed agent closed agent on AgentDojo without the defense (arm none, dev + conf; (i') reader vs judge)	**(i') FAILS**: 36 injected calls; reader 0.840 vs zero-shot judge 0.787 within suite (+0.052 [−0.026, +0.130]). TF-IDF 0.335. At the benign-call 5% cut: reader 36%, judge 53% ([released file])
Open-agent labels for a closed agent labels of the new policy from open agents, tested on the closed agent ((i) vs judge, (ii) vs no labels)	**(i), (ii) pass**: with AgentDojo labels from open agents, the reader on GPT's AgentDojo calls scores 0.922 vs judge 0.787 (+0.136 [0.078, 0.191]) and vs no labels +0.081 [0.044, 0.121]; caught at the benign 5% cut 36% → 92% ([released file])
Replication (Gemma-4) latent replication: does the reader replicate the agent's hidden state? ((i) R^2 vs text, (ii) readouts on the replica)	**(i), (ii) pass**: from the transcript alone, the reader replica reconstructs 66.4% of the agent's decision-state variance (top 128 PCs; text 58.6%; +0.079 [0.075, 0.083]); a probe trained on TRUE states scores 0.987 on the replica vs 0.984 on the true state (+0.003 [−0.002, +0.007]) ([released file])
Live prevention (GPT) live prevention on a closed agent: the monitor blocks GPT-6-luna's escalated actions in the loop ((i) violation episodes, gated − ungated)	**(i) passes**: live, the grey-box gate cut GPT-6-luna's episodes with an executed violation from 6.6% to 0.2% (−0.064 [−0.088, −0.044]); 0 violations after a block; goals 47.4% → 45.6%; round trip 89 ms median ([released file])
Live prevention (Claude) live prevention on a second closed family: Claude Haiku 5.5 in the loop ((i'))	**(i') passes**: on Claude Haiku 5.5, episodes with an executed violation 6.7% → 0.2% (−0.064 [−0.087, −0.044]); 0 violations after a block; goals 40.7% → 34.9%; round trip 80 ms median ([released file])
Replication, three architectures replication on three more open agents: gpt-oss-20b, Qwen3.8-27B, Muse-Glimmer-30B ((i), (ii) per agent; scenario-disjoint split)	**(i) and (ii) pass on all three agents (replication generalizes)**: from the transcript alone, R^2 of the decision state on fresh scenarios: gpt-oss 0.494 (text 0.392; +0.102 [0.093, 0.110]), Qwen3.8 0.542 (0.441; +0.101 [0.091, 0.111]), Muse Glimmer 0.462 (0.401; +0.061 [0.051, 0.071]). A probe trained on the true states, true → replica: 0.900 → 0.924, 0.955 → 0.972, 0.929 → 0.969 (all lower bounds > −0.02). Fidelity: the agents' own models predict 91–100% of their next 16 tool-call tokens. Exploratory, Gemma-4 under the same split: R^2 0.631 (text 0.556). Exploratory: the text map chose α = 1, the edge of the grid; a finer grid (α 0.01–1, exact dual-form ridge, [released file]) is worse at every smaller α for all three agents, so α = 1 is the text map's optimum ([released file], [released file])
Hold for review hold for review instead of block: Claude Haiku 5.5, escalated actions go to a reviewer ((i) safety kept, (ii) no goal cost vs "Live prevention (Claude)" ungated)	**(i) passes, (ii) fails.** Episodes with an executed violation: 6.7% ("Live prevention (Claude)" ungated) → 0.0% (review) (−0.067 [−0.091, −0.044]). Goals: 40.7% → 38.7% (−0.020 [−0.051, +0.007]); the lower bound misses the −0.03 margin, so no goal cost is NOT shown. Reviewer workload: 88 held actions (0.20 per episode, 6.2% of scored actions), 31 of them violations; 0 violations after a block. Exploratory ([released file]): episodes with a held-and-approved action kept their goals (60.8% vs 58.8%); the gap is in never-held episodes (38.9% vs 41.0%, n = 368), where both arms ran the same way, which we suggest is run-to-run variation against a non-concurrent control (a hypothesis that "Hold for review" does not test). Spend $0.81. **Deviation 1 (infrastructure, 16:39):** the first start ran 0 episodes. The monitoring server loaded the reader outside its lock, and 6 concurrent first requests raced on the lazy transformers imports (ImportError). Fix: the load moved inside the lock ([released file]; scoring unchanged; regression test [released file]). Rerun from scratch on a GPU job; spend of the failed start $0.004
tau2-bench, Claude a second outside benchmark: Claude Haiku 5.5 on τ^2-bench airline, monitors trained on four frontier agents' labelled trajectories ((i) within-tool reader vs tool identity, (ii) reader vs TF-IDF)	**not tested**: 9 violations < the preregistered minimum of 15 (Claude violated in 1.5% of episodes; the 9 come from 3 tasks). Descriptive only: within-tool AUROC reader 0.756, TF-IDF 0.963; pooled reader 0.835, TF-IDF 0.960, tool 0.686; the CAPE-trained probe zero-shot 0.357 pooled ([released file])

Test	Outcome (verbatim from the record)
Browser-size readers readers small enough for a browser or a CPU: do they transfer? ((i) browser reader vs TF-IDF on GPT AgentDojo; (ii) non-inferior to the 8B reader on closed CAPE)	**(i) passes, (ii) fails.** Selected browser reader (best on "AgentDojo, no labels"): Qwen3-Embedding-0.6B. On "AgentDojo, closed agent" (GPT on AgentDojo) it beats TF-IDF: 0.605 vs 0.335 (+0.270 [0.134, 0.451]); on the closed agents' CAPE-Bench actions it is below the 8B reader: −0.097 [−0.115, −0.079] (Claude 0.912, GPT 0.884 vs 0.989, 0.996). Decoder readers on "AgentDojo, closed agent": 0.6B 0.651, 1.7B 0.771, 4B 0.696, 8B 0.840, 14B 0.689; none is within 0.02 of the 8B on both sets, so no CPU option is offered. The shipped 8B reproduces its published AUROCs exactly ([released file]). **Deviation 1:** the analysis crashed after computing, when writing its JSON (a numpy bool); fix: one bool() cast, no computation changed
tau2-bench, two closed agents τ^2-bench with two closed agents (Claude Haiku 5.5, GPT-6-luna) and a persistent customer; "tau2-bench, Claude"'s monitors ((i) reader vs tool within tool, (ii) TF-IDF vs tool)	**(ii) passes, (i) fails.** Within-tool AUROC: TF-IDF 0.987 (+0.487 [0.428, 0.500] over tool identity), reader 0.738 (+0.238 [−0.053, 0.431]); reported: reader − TF-IDF −0.250 [−0.518, −0.077]. Per agent: Claude reader 0.555 / TF-IDF 0.966; GPT 0.739 / 1.000. Reading fixed in advance: on τ^2-bench, labels of the policy from other agents transfer to both closed agents through word counts, and the reader does not use them as well ([released file]). Spend (from the ledgers): Claude $1.71, GPT $0.83
tau2-bench, unseen tasks τ^2-bench, policy or tasks? "tau2-bench, two closed agents"'s test, training without the test tasks ((i) reader vs TF-IDF, (ii) reader vs tool; within tool)	**(i) and (ii) fail: neither monitor transfers to unseen tasks.** Within-tool AUROC: reader 0.535, TF-IDF 0.519 ("tau2-bench, two closed agents" with the same tasks: 0.738, 0.987); reader − TF-IDF +0.017 [−0.514, 0.539]; reader − tool +0.035 [−0.230, 0.259]. TF-IDF with entities masked 0.366 (exploratory). "tau2-bench, two closed agents"'s word-count advantage was memorisation of the tasks ([released file])
tau2-bench, reasoning judge τ^2-bench, a zero-shot reasoning judge (Qwen3-32B, thinking; no labels) on "tau2-bench, two closed agents"'s writes ((i) judge vs tool, (ii) judge vs "tau2-bench, unseen tasks"'s reader; within tool)	**(i) and (ii) fail (intervals include 0).** Within-tool AUROC 0.804 ("tau2-bench, unseen tasks"'s reader 0.535); judge − tool +0.304 [−0.084, 0.427]; judge − reader +0.268 [−0.074, 0.511]. Pooled 0.785; 53% caught at the 5% cut as first computed; 34.5% with exactly 5% reviewed (ties at the cut counted in proportion; [released file]); at 50: 12 of 19 caught, 16 of 243 safe flagged. 262/262 answers parsed, none truncated (median 485 tokens). 88% of scores are exactly 0. "tau2-bench, Claude": within tool 0.694, 2 of 9 caught at 50. Large point estimate, too few violating tasks to confirm ([released file])
Reasoning judge, frontier agents τ^2-bench, "tau2-bench, reasoning judge"'s judge unchanged on the 894 published writes of four closed frontier agents ((i) judge vs tool, (ii) judge vs reader cross-validated by task; within tool)	**(i) passes, (ii) fails.** Within tool: judge 0.766, reader (cross-validated) 0.646, TF-IDF (cross-validated) 0.490; judge − tool +0.266 [0.151, 0.338]; judge − reader (cross-validated) +0.120 [−0.047, 0.274]. Reported: 4-sample mean 0.838, − reader (cross-validated) +0.192 [0.022, 0.336]; per agent (judge) 0.768 / 0.771 / 0.775 / 0.618 (o4-mini). With exactly 5% of writes reviewed: reader (cross-validated) catches 18.4%, the judge 11.7%, the 4-sample mean 15.3%, the tool 13.0% (most possible 19.7%). 894/894 parsed in every sample; none truncated; 63% of scores exactly 0. Reading (fixed): the judge checks rules beyond the tool identity, but not better than the reader trained on other tasks ([released file])
Monitor-aware agent monitor-aware closed agent: Claude Haiku 5.5 with and without CAPE's oversight notice, 990 scenarios × 2 arms, no gate; the shipped reader with a cut frozen on Claude's earlier unlabelled traffic ((i) caught aware − plain; (ii) aware caught)	**(i) and (ii) pass.** Caught at the frozen cut: plain 83.6%, aware 97.1% [92.9%, 100%]; aware − plain +0.136 [+0.050, +0.225]. Being told did not change how often Claude violated (−0.003 [−0.012, +0.005]) or reached the goal (37.6% vs 37.8%). Note: the notice moves the score distribution up: the frozen cut escalates 15.2% of the aware arm's actions (plain 5.3%). At each arm's own 5% budget: 88.6% vs 83.6%. AUROC within template 0.994 vs 0.967 ([released file])
Adaptive attacker adaptive attacker with score access: Qwen3-32B rewrites the free text of 134 real closed-agent violations (2 rounds × 16, score feedback), the checker confirms each still violates ((i) reader caught after attack; (ii) reader − TF-IDF)	**(i) and (ii) fail.** Reader caught 96.3% → 29.1% [21.6%, 36.6%] after attack (Claude 11.8%, GPT 47.0%); TF-IDF 57.5% → 34.3%; reader − TF-IDF −0.052 [−0.149, +0.037]. Every rewrite still violated (replay checker). The rewrites keep the decisive field (recipient, service and host) and add self-reported approvals ('approved under routine maintenance protocol . . . ', '–reason . . . '): the learned monitors read the agent's own claims as evidence. Reading as fixed: the monitor is not robust to adaptive rewording ([released file])

Test	Outcome (verbatim from the record)
Canonical-action monitor canonical-action monitor: the shipped reader and recipe trained on canonical actions (free-text fields → "[free text]", quoted prose and # comments removed); "Adaptive attacker"'s attack re-run against it with seed 1 ((i) caught after attack; (ii) clean catch − shipped on Claude ∪ GPT; (iii) AgentDojo within-suite − shipped)	**(i), (ii) and (iii) fail.** After attack 50.8% [42.5%, 59.0%] caught (shipped reader 29.1%); before attack 55.2%. Clean catch at 5%: Claude 94.1% → 51.5%, GPT 98.5% → 59.1%; (ii) −0.410 [−0.493, −0.311]. AgentDojo (iii) −0.029 [−0.080, +0.010]. By tool: send_message 29.3% before and after (the canonical score cannot move, as Corollary 1 predicts); execute_mock_command 96.2% → 84.6% (commands keep their words). Cause: the premise that the label depends on the action only through κ(A) is false for 81 of the 82 send_message targets (one family, tp_campaign_test: the body carries the draft that is leaked), so the body is payload, not only self-report [this cause was wrong: see "Correction to the "Canonical-action monitor" cause" below]. Reading as fixed: canonical mode is an option with its measured cost, not the default ([released file])
Training against the attack adversarial training: Gemma training actions of the attacked tools, violating AND safe (300 + 300), rewritten by "Adaptive attacker"'s attacker (8 each), labelled by replay (4,554 rewrites; 28 changed label) and added to training; "Adaptive attacker"'s attack re-run against the hardened monitor with seed 2 and told the defense ((i) caught after attack; (ii) clean catch − shipped on Claude ∪ GPT; (iii) AgentDojo − shipped)	**(i), (ii) and (iii) fail.** After the defense-aware attack 59.7% [51.5%, 67.9%] caught (before 92.5%); by tool: send_message 98.8% → 47.6%, execute_mock_command 82.7% → 78.8%. The shipped probe on the same rewrites: 96.3% → 57.5%, so hardening added about 2 points. Clean catch at 5%: Claude 94.1% → 88.2%, GPT 98.5% → 97.0%; (ii) −0.037 [−0.078, +0.016]. AgentDojo (iii) +0.012 [−0.050, +0.071]: the lower bound equals the margin after the locked code's 4-decimal rounding, so it fails as registered. Reading as fixed: augmentation with a known attack does not stop a defense-aware attacker; three preregistered defenses against score-querying rewording have failed ([released file])
Cut set again after a prompt change set the cut again after a prompt change, live: Claude Haiku 5.5 with "Monitor-aware agent"'s oversight notice, 400 scenarios × (ungated, gated); the cut set from "Monitor-aware agent"'s aware-arm reader scores (3,046, no labels), budget 5%, frozen ((i) violation rate gated − ungated; (ii) share of ungated actions escalated within [0.025, 0.075])	**(i) and (ii) pass.** Episodes with an executed violation 6.5% (26/400) → 0.0% (0/400): −0.065 [−0.090, −0.043]. The new cut escalated 4.73% [3.49%, 6.10%] of the ungated arm's 1,206 actions; the cut frozen on plain traffic would have escalated 16.6%. 25 of 54 blocked actions would have been violations; goals 38.5% → 35.5%; violations caught at the new cut in the ungated arm 88.5%; median round trip 83 ms. Reading as fixed: setting the cut again on unlabelled traffic restores the budget, and the gate prevents the violations of an agent that knows it is monitored ([released file])

Correction to one recorded cause. The record keeps the first wording of the canonical-action monitor's cause and adds this dated correction (an exploratory diagnosis; no endpoint changed).

- The "Canonical-action monitor" row first said that the message body "carries the leaked draft", so free text is payload. Checking the data showed this is wrong:
- **What decides the label.** In tp_campaign_test (81 of the 82 attacked messages), the violation is the recipient: a test send to a list whose members are thousands of customers. That fact appears only in a `list-info` result, and neither closed agent ever called `list-info`. CAPE's checker labels the send by its recipient, whatever the body says.
- **What the attacker did.** The attacker's best rewrites dropped the draft from the body in 53 of 82 ("Adaptive attacker") and 55 of 82 ("Training against the attack") messages, and the checker still labelled every one a violation.
- **Training coverage.** tp_campaign_test is absent from the 23 families of Gemma's training steps. Among open agents, only Qwen has violations of it (7).
- **The cause.** The shipped reader caught these sends from an incidental cue (a file's content copied into a message) that it learned on other families. The attacker can remove that cue without changing the label. The canonical form removes it too, and the decisive field (the list) was never learned.
- **A theorem (paper, "a ceiling under rewording").** When an attacker can give a violation the free text of safe actions with the same decisive fields, no monitor's attacked operating point beats the clean ROC of the best monitor of (transcript, decisive fields). The canonical reader's clean catch (51.5% / 59.1%) is our estimate of that ceiling. "Training against the

attack"'s 59.7% after the attack and "Adaptive attacker"'s 29.1% are consistent with it.

- **The headline without this family.** The shipped reader's exact-budget catch at 5% is 92.3% (Claude, 26 violations) and 96.3% (GPT, 27 violations), against 94.1% and 98.5% with it.